

Project 3 – Interactive Visualization Write Up

Motivation

Our motivation is to explore how different emissions futures may affect projected temperature change across the United States within the next decade. Instead of showing raw temperature, we focus on the change from an observed 2025 baseline, so users can compare projected warming or cooling from a common starting point.

We compare three pathways: Low, Medium, and High emissions, based on the SSP scenario framework used in CMIP6 and IPCC (Intergovernmental Panel on Climate Change) climate projections. The guided story lets users follow one pathway from 2026 to 2035 and see where warming stands out. Because each pathway leads to its own finding, users may see a different takeaway depending on whether they choose Low, Medium, or High emissions. After the story, users can explore different years, pathways, and states on their own. Overall, our goal is to show that projected warming is not evenly distributed, and that different emissions futures can lead to different spatial patterns, especially across the central U.S.

As background context, the guided story also explains why temperature change matters: even small increases in global temperature can raise climate risks and make extreme events more severe, according to the United Nations Climate Action.

Design Rationale

Since our main question is spatial, we chose a U.S. choropleth map as the main view. This lets users see where projected warming or cooling is concentrated across states and counties.

Before mapping the data, we applied a delta-change transformation. For each year and emissions pathway, we calculated the temperature difference between the CMIP6 projected temperature and the model's own 2025 prediction. We then added that change to the observed 2025 baseline temperature. The mapped value is the final change from the observed 2025 baseline, which lets us show projected change relative to a real starting point instead of using raw model temperatures directly.

We encoded temperature change with a diverging blue-to-red color scale centered at zero. Blue represents cooling from the 2025 baseline, red represents warming, and the middle represents little change. We fixed the displayed range at -3.5°C to $+3.5^{\circ}\text{C}$ so colors stay comparable across years and emissions pathways. Values beyond the endpoints use the endpoint colors, so extreme values do not dominate the map. We also used an orange outline to highlight the selected or story-focused state.

For interaction, we combined a guided storyline with open exploration. The story introduces the baseline, asks users to choose an emissions pathway, animates the map from 2026 to 2035, and highlights where warming stands out. Since each emissions pathway has its own guided result, users may see a different finding depending on whether they choose Low, Medium, or High emissions. The year and pathway controls then let users explore how projected warming changes under different assumptions. Hover tooltips show Low, Medium, and High emissions values for the selected year, and click-to-zoom lets users move from state-level to county-level detail.

We considered using line charts to compare pathways over time. While line charts would make temporal comparison easier, they would hide the spatial pattern that is central to our research question. Since our main goal is to show where warming stands out, we chose a map-based visualization.

Development Process

In terms of the development process, we divided the project into several parts. Huiyu and Song worked mainly on data cleaning and shaping the storyline, including the decision to compare projected temperatures against the 2025 observed baseline. The implementation and debugging process was shared across all three members: Huiyu, Song, and Ricki. Huiyu and Ricki worked on the final write-up.

We estimate that each of us spent roughly around 30+ hours on the project, so in total, roughly 90+ people-hours. We spent most of the time designing the storyline logic and debugging. Since the visualization combines a guided story with interactive controls, we had to make sure the selected year, emission pathway, map zoom, tooltip, and reset behavior were all updated consistently.

References

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<https://www.dkrz.de/en/communication/climate-simulations/cmip6-en/the-ssp-scenarios>

Statista. “Annual CO₂ emissions worldwide from 1940 to 2025.”
<https://www.statista.com/statistics/276629/global-co2-emissions/>

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